Time dilation from counting events without pre-assigned places, and the Feynman checkerboard up to a moving atomic clock: registered lattice tests

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

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Abstract

This record reports a sequence of lattice tests motivated by four counting premises and implemented through known physical constructions, with the further assumptions stated explicitly. With a light-cone order and a clock that reads the longest chain of events, independent uniform random events recover proper time within the reported finite-size accuracy, while a synchronous grid gives a chain length that does not depend on the velocity; a clock that counts events and multiplies the two light directions recovers the Lorentz factor on that same grid, through the exact identity $\sqrt{(ab)}/((a + b)/2) = \sqrt{(1 - v^2)}$. With an imaginary weight for every change of light direction the count becomes the Feynman checkerboard, a Dirac quantum walk; the subsequent tests concern mass as the turning angle, fermion doubling with net chirality zero, boundary (domain-wall) fermions, the gauge field of MM 2.27 on the same lattice (Landau levels, Maxwell's evolution equations with Coulomb's law in 2D and 3D, photons with two polarizations, a fitted Casimir energy coefficient of 0.9997 times $-\pi^2/720$), the Schwinger model in 1+1, and hydrogen-like atoms: for a Bohr radius of six lattice steps the emission rates agree within 1.7 % with the nonrelativistic hydrogenic formula at the same coupling and effective mass.

Time that slows down, counted with events: from the Feynman checkerboard to a moving atomic clock, with registered lattice tests

Student version of MM record 2.40.2 (DOI 10.5281/zenodo.23631239), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

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What version 2.40.2 adds (25 October 2026).

- It adds section 3.1. With more than two counted sides, the product of all the sides is not *Lorentzian*, that is, it does not respect relativity. The product taken pair by pair, instead, is. And n sides give n − 1 directions of space: our 3+1 spacetime corresponds to four sides.
- It adds to section 13 a precedent not cited before: Knuth and Bahreyni (2014). It also adds a 2016 report by Knuth on the number of chains.
- It states in section 12 that the number of dimensions of space is not derived.
- Everything that belongs to version 2.40.1 stays unchanged: programs, outputs, registered tests and their totals.

What version 2.40.1 corrected (15 October 2026). It corrected two statements of version 2.40 (10 October 2026).

- The massive palindromic automaton has 64 Dirac valleys, not 16 species (section 5).
- With the parameters used, the 3D hand on the boundary also has boundary branches near half period (section 6).

The original programs, the outputs and the totals of the registered predictions have not changed. The explanatory text has

changed, and extra checks have been added.

The framework. MM is the author's framework of ideas. In MM physical quantities are *read* from counts. This deposit uses only the small part of it stated in section 2. Ugo Lanciano is not a professional physicist. The calculations were done with an AI assistant.

Before starting: the words you need

Event. Something that happens at one point and at one instant: a flash, a tick of a clock.

Spacetime and light cones. Space and time together form *spacetime*. Nothing goes faster than light. From an event, light spreads out in all directions: it draws a *light cone*. An event can influence another one only if that one is inside its light cone. This creates an *order*: what comes before and what comes after.

Proper time and time dilation. *Proper time* is the time shown by a clock that travels with an object. According to Einstein's relativity, a moving clock runs slower than one at rest. The factor is $\sqrt{(1 - v^2)}$, where v is the speed divided by that of light. This slowing is called *time dilation*. Its inverse, $1/\sqrt{(1 - v^2)}$, is called the *Lorentz factor*, γ.

Lattice and ticks. A *lattice* is a grid of points, like a checkerboard. Here space and time are made of steps: the lattice step is a = 1. At each *tick* light takes one step.

Arithmetic mean and geometric mean. The arithmetic mean of two numbers is their sum divided by two. The geometric mean is the square root of their product. An example with numbers chosen by us: for 1 and 9 the arithmetic mean is 5, the geometric mean is 3. The arithmetic mean is never smaller than the geometric one.

Amplitude and imaginary weight. In quantum mechanics every way in which something can happen has an *amplitude*: a number that one can think of as an arrow, with a length and a direction. An *imaginary* weight turns the arrow instead of lengthening or shortening it. Probabilities come from the length of the final arrow.

Spin and hand. *Spin* is a quantum property of particles. It resembles a small rotation on itself, but it is not a little ball that spins. *Chirality*, or *hand*, says whether a particle is *right-handed* or *left-handed*, like our two hands: equal but not superimposable.

Fermions, photons, atoms. *Fermions* are the particles of matter, like the electron. *Photons* are the particles of light. In an atom the electron sits on energy levels, called 1s, 2s, 2p. When it drops from a higher level to a lower one, it emits a photon.

Registering a prediction. It means writing it down and dating it before doing the calculation. In this way it cannot be changed after seeing the result.

In short

Time from counting.

- A light-cone order is assumed. One uses a clock that reads the *longest chain* of events, that is, the longest row of events, each inside the light cone of the previous one.
- With events placed at random, each independent of the others, one recovers the proper time of special relativity.
- With a *synchronous grid*, that is, with places fixed in advance, the length of the chain does not depend on the velocity.
- A clock that counts the events and multiplies the two directions of light instead recovers the Lorentz factor, even on that grid.

From counting to the Feynman checkerboard. Every time a history changes light direction, it is given an imaginary weight. Then the count becomes the *Feynman checkerboard*, a *Dirac quantum walk*.

The field and the atom. On the lattice one adds the gauge field of MM 2.27. One tests, one step at a time: Landau levels, Maxwell's evolution equations, photons, the Casimir energy and a hydrogen-like atom. The atom's emission rates of light agree within 1.7% with the hydrogenic formula, at the same coupling.

The moving atom. At the lattice spacing that could be computed, the moving atom does *not* show Einstein's slowdown of its internal frequency. If the free dispersions of the lattice are replaced with those of the continuum, the coefficient

becomes −0.44 (−0.49 in the strictly first-order ratio). Einstein predicts −0.50.

What is new and what is not. All the structures are known physics. The deposit puts them together as a sequence, motivated by four premises on counting, with the other assumptions declared. Every prediction had its fingerprint before the calculation. Those that failed are reported together with the others.

More than two sides. With more than two counted sides, only the product taken pair by pair keeps the Lorentz factor (section 3.1).

1. What is known, and what is proposed

Everything has a known counterpart. Every result below already exists in physics:

- the longest chain of events distributed at random (a *Poisson distribution*) as proper time, in *causal sets*;
- the Feynman checkerboard and the Dirac quantum walks;
- Wilson and domain-wall fermions;
- lattice electrodynamics;
- the Schwinger model;
- the *golden rule* for the spontaneous emission of light;
- the Breit interaction.

Section 13 lists the sources and what is different here.

What the deposit proposes. A single sequence that starts from counts and reaches a moving atomic clock:

1. **The clock.** A clock that counts events without places fixed beforehand runs at the geometric mean of the two light sides: this is Einstein's proper time. A clock that reads the longest chain on a grid of places gives the arithmetic mean: this is Galilean time, without dilation (section 3). With more than two counted sides, only the pairwise product keeps the Lorentz factor, and n sides give n − 1 directions of space (section 3.1, added in 2.40.2).
2. **The checkerboard.** The same count, with an imaginary weight for each turn, is the checkerboard: a Dirac quantum walk. In the three-dimensional construction used here spin ½ is required. Mass is the turning angle (sections 4-5).
3. **The hand on the boundary.** Massless fermions with a single hand live on the boundary between two regions of opposite turning. Two boundaries bind into a mass that falls with their distance (section 6).
4. **The field.** The field of MM 2.27, put on the same lattice, gives gauge invariance, Maxwell's evolution equations, photons and an atom that emits light (sections 7-9).
5. **The moving atom.** On the lattice computed here its frequency does not slow down. With the dispersions of the continuum it does, close to Einstein's coefficient (section 10).

The sequence is tested, not proved like a theorem. The four premises of section 2 motivate it, but alone they do not determine it. Other assumptions are needed, declared where they are used: the light-cone order, the Dirac matrices and the order of the steps, the Wilson term, an extra direction for the boundaries, the field equations of MM 2.27, canonical quantization, the forms of the interactions and the *no-pair* projection. Each step is a numbered test with predictions registered beforehand.

2. The model: four premises and the declared choices

The units. The lattice step is a = 1. The speed of light is c = 1. The reduced Planck constant is ħ = 1. Light moves one step per tick. A light-cone order is assumed.

The premises of the framework, in technical form.

- **A (no place assigned beforehand).** Events ("copies") are not placed on a set of places given in advance. In the tests they are taken as independent and distributed at random, uniformly, in the volume of spacetime. This is an assumption added to A: A motivates it, but does not imply it.
- **B (second link).** In an elementary step the two counts a and b of the light sides of a clock are *coprime*, that is, they have no common divisors other than 1. The copies arise at their common multiples. So a light diamond of sides a and b contains ab = gcd(a, b)·lcm(a, b) cells.
- **C (opposite directions multiply).** A clock reads the number of copies and combines the two opposite light directions by product. Its time is √(ab).

- **D (histories that never happened).** Each change of light direction of a history (a "turn") carries an imaginary weight. The amplitude of an outcome is the sum over all possible histories.

The gauge field is that of MM 2.27: phases on the links of the lattice, the equations of its script F, with $\alpha = \beta = 1$.

The declared choices. Each one is declared where it is used.

- **The Wilson term r.** The turn depends on the neighbours. In this way the extra species of the cubic lattice become heavy (section 5: 63 extra Dirac valleys of the massive automaton, corrected in 2.40.1).
- **The palindromic order** of the axes within a tick: the order of the steps along the axes reads the same in both directions, like a palindromic word. It removes a spin splitting that should not be there.
- **The regularized Coulomb potential** $-Z\alpha/\sqrt{(r^2 + 0.25)}$, which does not become infinite at the centre.
- **The continuous-time form of the checkerboard**: the lattice Dirac Hamiltonian $H = \Sigma_a \alpha_a \sin p_a + \beta(m + r \Sigma_a(1 - \cos p_a))$. It is used for the electrodynamics in 1+1 (section 8) and for the atoms (sections 9-10). Sections 4-7 use the discrete-time automaton itself. Section 3 only counts events.

What the framework does not fix.

- The value of a single mass. The turning angle θ is a free parameter, as m is in the Dirac equation.
- The lattice step in physical units. If it were the Planck length, the electron would accumulate one radian of mass phase in about 2.4×10^{22} ticks.

3. The clock (P1-P7)

The light diamond. A clock moves at speed v. Between the start event and the end event, light rays draw a diamond. Its sides are $a = \Delta t - \Delta x$ and $b = \Delta t + \Delta x$.

arrangement of copies	what the clock reads	result
uniform at random, 1+1, 200,000 copies	longest chain	$\sqrt{(1 - v^2)}$ within +0.2 to −0.9% for v = 0.3-0.9 (within 0.3% after a finite-size correction)
uniform at random, 3+1, about 6000 copies	longest chain	moving and resting clocks with the same proper time agree within 1.3 σ; the fixed tolerance of 6% failed at v = 0.8 (about 780 copies)
synchronous grid	longest chain	(a + b)/2 for every v: a chain length that does not depend on the velocity
fixed places, random times	longest chain	dilation 7-29% stronger than Lorentz: the places define a frame
synchronous grid	number of copies, product rule (B, C)	$\sqrt{(1 - v^2)}$ within 0.43%
light lattice	$\sqrt{(ab)}$ against (a + b)/2	$\sqrt{(1 - v^2)}$ to 3×10^{-16}

How to read it.

- With copies placed at random, the clock that reads the longest chain recovers Einstein's dilation, $\sqrt{(1 - v^2)}$.
- In 3+1 (three dimensions of space and one of time) the agreement is within 1.3 σ. But at v = 0.8 the fixed tolerance of 6% is not respected: there are only about 780 copies there.
- On a synchronous grid the longest chain gives (a + b)/2 at every speed: no dilation, as in Galilean time.
- With fixed places and random times the dilation is 7-29% stronger than Lorentz's: the fixed places create a privileged frame.
- On the synchronous grid, a clock that counts the copies with the product rule instead recovers $\sqrt{(1 - v^2)}$ within 0.43%.

The reason in one line.

- The number of copies in the diamond is proportional to ab.
- A clock that reads $\sqrt{(ab)}$ runs at the geometric mean of the two sides. A sum of the sides instead gives the arithmetic mean.
- **The arithmetic mean exceeds the geometric mean, except at rest. The gap is time dilation:** $\sqrt{(ab)}/((a + b)/2)$

$= \sqrt{(1 - v^2)}$. It is an exact identity.

The step at rest.

- A clock at rest takes steps of (1, 1).
- A step with coprime sides is *indivisible*: the segment from (0, 0) to (a, b) has gcd(a, b) − 1 lattice points inside it.
- If the sides were different in every step, a clock at rest would zigzag. It would tick like a straight clock at c/3 (P7c). This is why the step at rest is (1, 1).

A registered criterion not met.

- For P1-P5 as a whole this criterion was registered: "if P1 or P5 fails, the right angle does not follow from the count".
- The criterion was not met, because P5 failed its 6% tolerance at v = 0.8.
- The comparison of P5 at equal proper time passed. Reading the failure as a finite-size effect is an interpretation made afterwards.
- The same tests, with the same programs, are section 10 of MM 2.36.1. There they are called M1-M7, and premises A and B are called P7 and P6.

3.1 More than two sides: the pairwise rule (added in 2.40.2)

The question. Premise C multiplies the two light sides of a clock. What if a clock were described by n independent counts, $x_1, \ldots, x_n$? (For n = 2 they are a and b.) Premise C can be extended in two ways, which coincide for n = 2.

- **The product rule.** The clock reads the geometric mean $(x_1 \cdots x_n)^{(1/n)}$, with an order in which every count increases.
- **The pairwise rule.** Each pair of sides is multiplied, and the pairs are added: $Q = \Sigma_{i<j}\, x_i x_j$. The clock reads $\sqrt{Q}$, with an order in which the total $T = \Sigma\, x_i$ increases and Q of the difference is positive.

What is found.

- **Q has the form of relativity.** It has signature (1, n − 1) and is equal to $((n - 1)/(2n))T^2 - \tfrac{1}{2}\Sigma(x_i - \bar{x})^2$. There is one direction of time, the common growth of the sides. There are n − 1 directions of space, their spread. In these directions the form is *isotropic*, that is, the same in all directions (swapping the sides does not change it). Three sides give 2+1, four give 3+1; for n = 2, Q = ab. It holds for every n: it follows from the Gram matrix of the sides.
- **The product does not have relativity.** For $n \geq 3$, the only diagonal transformation with positive factors that preserves Q is the identity ($\lambda_i \lambda_j = 1$ for every pair forces $\lambda_i = 1$). The product instead is preserved by rescalings with $\prod \lambda_i = 1$ and by swaps of the sides, but not by rotations: it has no Lorentz group.
- **The test on counted structures.** Structures are built from independent rank counts, without putting them in Minkowski coordinates.
- With four sides and the pairwise rule, the *Myrheim-Meyer dimension* (a way of estimating the dimension from a structure of events) is 3.92 ± 0.11. It is measured on the six sampled intervals estimated to lie at least 95% inside the region. On all the sampled intervals it comes out as 3.61.
- The moving clocks of the first observer give ratios of 0.981 and 0.998 (means over 10 seeds, in the classes v = 0.2-0.4 and 0.4-0.6).
- With three sides one gets 2.98 ± 0.05, and ratios 0.977 and 0.994.
- With the product rule, the same four counts give a structure of dimension about 3.7, whose moving clocks are off by 12% (1.121).
- A control with Poisson copies in 3+1, measured by the same code, gives 0.992 and 1.024.

What is known and what it means.

- The algebra is known: Q is the Minkowski interval along n light directions with equal mutual products (section 13).
- The pairwise rule defines, for every n, a Lorentzian form with n − 1 directions of space.
- The tests on the rank structures cover three and four sides. There the pairwise rule holds, the product rule does not.
- The number of sides is not derived (section 12).

The limits.

- The registered comparison between two observers, in the test of the moving clocks, failed for both pairwise structures: −3.4% and −7.5% in the fastest class. That form of the test is not validated beyond 1+1. This is why only the first observer is used above.
- The estimator reads 3.61 on all the sampled intervals, because the region of the ranks is not *causally convex* for this order. A registered diagnostic separates the intervals estimated to lie inside the region.
- The registered prediction for the dimension from the longest chain also failed (3.45). That diagnostic does not concern it.
- The isotropy of the sampled structures was not tested: only that of the form was.
- **The count of the tests of the addition:** 20 passed, 4 failed, 4 not evaluable or incomplete.

4. The checkerboard (P8-P9)

In one dimension (1+1). At each tick a history takes a step along one of the two light directions. Each turn carries an imaginary weight of size θ = εm (premise D). The registration writes it +iεm, the code −iθ: it is only a sign convention. All the histories are summed (Feynman and Hibbs 1965, problem 2-6). One finds:

- The normalized amplitude A tends to $J_0(m\tau)$, a known function (the Bessel function). The error is 3.5×10^{-3} at εm = 0.002, and it halves (ratio 2.01) when the step is halved.
- The other component carries the factor $\sqrt{((1 + v)/(1 - v))}$ in front of J_1, as a *spinor* component must (the mathematical object that describes particles with spin ½).
- A depends only on the proper time τ, up to v = 0.95, within 7.7×10^{-3}. This is Lorentz invariance, in the limit of fine steps.
- The first zero of the moving clock is delayed by γ within 0.06%.
- **With real weights instead**, that is, with Kac's classical zigzag, the sum depends on the laboratory time: 0.4607 against e^(−aτ(γ − 1)) = 0.4614. **It is the imaginary weight that makes the count Lorentz invariant.**

In three dimensions (3+1). At each tick the history takes a step along x, y and z, each in one of the two light directions of that axis, chosen by the spin. Then the turn exchanges the chiralities. The four matrices *anticommute*: their squares add without cross terms. One finds:

- $\omega^2 = k_x^2 + k_y^2 + k_z^2 + m^2$ within 1.4×10^{-3}. It is the relation between energy and momentum in relativity.
- The group velocity is right within 3.8×10^{-4}, and it is parallel to k.
- The rate of the clock is $m\sqrt{(1 - v^2)}$ within 9.4×10^{-4}.
- The error is divided by 4.00 when the step is halved.
- Without the palindromic order, the error rises to 4.3% on the diagonal of a face at 0.9 c (estimated spin splitting 4.26%).
- Without spin, a specific zigzag without spin on six directions has three modes at rest that do not propagate. **In this construction spin ½ is required.**

It is a member of the family of Dirac quantum walks, with the step split into several parts (Bialynicki-Birula 1994; Meyer 1996; Arrighi, Nesme and Forets 2014; D'Ariano and Perinotti 2014).

5. Mass, and relations among particles (P10-P11)

- **Mass is the turning angle per tick.** The rest frequency is equal to θ exactly (error $\leq 2 \times 10^{-16}$).
- **In the one-dimensional automaton the band becomes flat at θ = π/2** ($|\partial\omega/\partial k| \leq 6 \times 10^{-14}$). It means a turn at every tick and a particle that cannot move. It is a property of that automaton ($\cos\omega = \cos\theta \cos k$), not a general limit. In the palindromic 3D automaton the band at θ = π/2 is not flat.
- **Rest mass and inertial mass coincide only for small θ.** In 1D, inertial/rest − 1 = θ²/3: 3.36×10^{-5} against 3.33×10^{-5} at θ = 0.01, in a recheck with a finer derivative step. In the 3D automaton it is about 4θ²/3 (1.34 θ²).
- **Particle and antiparticle** have the same mass: the spectra are symmetric to 2×10^{-15}.
- **Free fall.** A potential stretches the local step, as the *lapse* of MM 2.36 does. Then a = −gθ/tan θ within 0.12%. Light particles (θ ≪ 1) fall in the same way. The correction due to the finite step between two masses, $(\theta_1^2 - \theta_2^2)/3 \approx (m/m_P)^2/3$, depends on the mass.
- **Hands come in pairs.**
- The massless (Weyl) automaton has 16 *nodes*: 8 at $\{0, \pi\}^3$ with chirality +1 and 8 at $\{\pi/2, 3\pi/2\}^3$ with chirality −1. The net chirality is zero.
- No small local turn opens a *gap*, that is, a jump in energy (10 cases out of 10).
- The massive automaton (palindromic, two-tick) has 64 Dirac valleys with gap θ in the Brillouin cube $[0, 2\pi)^3$: 32 around quasi-energy 0 and 32 around π/2. They are 8 in each of the eight parity sectors conserved by the two-tick evolution without the Wilson term.
- **Correction of 2.40.1:** version 2.40 said 16 species: it had checked 16 of the 64.

- This is consistent with *fermion doubling* and with the Nielsen-Ninomiya theorem, whose hypotheses concern static Hamiltonians.

The framework does not fix the value of θ for any particle. It fixes only the relations above.

6. The hand on the boundary (P12-P15)

The idea. In two regions the turn has opposite sign: $\theta(s) = \pm\theta_0$, with $\theta_0 = 0.2$, and there is the Wilson term. Then massless states with a single hand live on the boundaries.

- **2D strip, r = 0.3.** One mode per boundary, with slopes +1.00000 and −1.00000 and $\langle \sigma_x \rangle = \pm 1.0000$. This holds in a recheck with a k grid shifted off k = 0: there the registered detector had failed. Without the Wilson term there are four right-handed and four left-handed modes per boundary.
- **3D (with a fourth direction s across the boundaries), r = 0.15.**
- Near quasi-energy 0 there are two states per boundary with $\omega = \pm|k|$ ($\omega/|k| = 1.0000$), helicity ±1.000, chirality +1 on one boundary and −1 on the other.
- Near quasi-energy 0 there are no boundary states in the sampled neighbourhoods of the other 15 special points.
- With r = 0 there are 60 extra states per boundary.
- **Near half period (quasi-energy π/2 per tick; added in 2.40.1).**
- There are extra boundary branches, called *temporal doublers* of the hand on the boundary.
- On a sampled 24^3 grid, light states inside the gap appear near π/2 only at k = 0 (8 states) and at the six points $(\pm\pi/2)e_j$ (4 each).
- They are consistent with a criterion on the sign of the mass: in these valleys the Wilson mass (r) is smaller than θ_0, so the effective mass changes sign across the boundary.
- Their mean distance from π/2, at the separation used here, is 0.0127 at k = 0 (0.0122 at $(\pi/2)e_x$). It decreases approximately exponentially with the separation, at a rate close to 0.05 per site.
- In a control with $\theta_0 = 0.1 < r$ they are absent.
- **Two boundaries at distance d bind into a particle with a true mass** ($\omega^2 - k^2 = m^2$ within 3.5%), shared half and half between the two boundaries. The mass falls nearly exponentially with d: m/θ_0 goes from 1.0×10^{-2} at d = 10 to 1.9×10^{-4} at d = 35, in 3D. The local rate is 0.13-0.18 per site and grows with d. It is the same as in 2D within 1.4%.

A small mass does not need a small number: it needs distance. The localization goes back to Jackiw and Rebbi (1976). The chiral lattice construction is the domain-wall fermion of Kaplan (1992) and Shamir (1993), here inside the checkerboard.

7. The field of MM 2.27 on the checkerboard (P16-P19)

The gauge field as phases on the links.

- The spectrum does not change under local phase changes (1.5×10^{-14}). This is *gauge invariance*.
- In a magnetic field the levels are $\pm\sqrt{(m^2 + 2nB)}$ within −1.0 to +1.6% for n = 1-3, with the n = 0 level on one side only. These are the *Landau levels*.
- In an electric field, over the tested range $p \le 3\theta$, the speed stays below 1 (maximum 0.945). It approaches the continuum value $v = p/\sqrt{(p^2 + \theta^2)}$. At switch-on there are oscillations (*Zitterbewegung*, a trembling, with period about 32 ticks). These oscillations made the registered 2% tolerance fail.

The dynamic field in 2D.

- The evolution equations of MM 2.27 are used (φ on the nodes, A on the links, a jumping method called *leapfrog* with dt = 1/2), with the current of the checkerboard as source.
- The coupling is staggered by half a tick, so Gauss's law is not exact, as already in MM 2.27.
- Charge is conserved to 8.7×10^{-18}.
- The field of a charge reaches distance 24, with its maximum at t = 28.
- Like charges repel and opposite charges attract, with nearly equal and opposite displacements: ratio 1.037 on a 192 × 192 torus. This is a recheck, made after the registered 96 × 96 torus proved too small.

The dynamic field in 3D.

- Charge is conserved to 3.5×10^{-18}.
- At R = 20 the field reaches 1% at t = 11 and has its maximum at t = 21.
- Coulomb's law: $E \cdot 4\pi r^2/Q = 0.988, 0.990, 0.970, 1.077$ at r = 10, 14, 18, 22.

- Attraction and repulsion are symmetric (ratio 1.001).

The quantized free field in 3D.

- Two polarizations (rank 2 to 1.4×10^{-14}), as for real light.
- $\omega/|k|$ is right within 0.36% for $|k| \le 0.3$.
- The *black-body* energy is within 2.5% of the Stefan-Boltzmann law at T = 0.1.
- **The fitted coefficient of the Casimir energy is 0.9997 of $-\pi^2/720$**, that is, right to 0.03%. The *Casimir effect* is the attraction between two close plates, due to the quantum vacuum. The energy per area is C/d^3; the force is $3C/d^4$.
- The zero-point energy is 2.39 per cell.

These are numerical checks of the evolution equations adopted. They do not check the full set of Maxwell's constraints.

8. Quantized field and matter in one dimension, 1+1 (P20)

The calculation. An exact diagonalization, up to 184,756 states, of the lattice *Schwinger model*. It is electrodynamics in a single dimension of space. The flux on the links is quantized by Gauss's law, and matter sits on staggered sites (Kogut and Susskind 1975).

- **Confinement.** The energy between two static charges grows as $g^2/2$ per link: slope 0.4974 against 0.5.
- **String breaking.** With light matter, the vacuum creates a pair of particles that screens the charges: the slope goes from 0.290 to 0.022, and the induced charge is −0.993.
- **The Schwinger mass is not reached at these sizes:** 0.732 against $1/\sqrt{\pi} = 0.564$ at N = 20. It decreases slowly with N: 0.760, 0.744, 0.732 for N = 16, 18, 20. At these chain lengths the continuum limit is not established.

9. An atom that emits light (P21-P22)

The atom on the lattice. The electron is the continuous-time checkerboard with r = 0.2. The nucleus is a fixed charge. The photons are the quantized modes of section 7, coupled through the phases of the links with the physical value $\alpha = 1/137.036$. The emission rates come from the golden rule, at the lowest order.

The small atom (64^3 torus, m = 0.8, $Z\alpha = 0.6$, Bohr radius 2.4 steps).

- 1s, 2s and 2p exist, with the fine structure in Dirac's order.
- The coupling on the lattice is equal to the dipole form in the long-wavelength limit (0.999999).
- **All four registered predictions failed.** In the registered run the 2p-1s frequency is 75% above the continuum. The rates are 34-44% below the hydrogenic formula (−32% and −46% after a recheck that re-diagonalized H in the odd sector).
- The atom is too small. The 1s feels the lattice, and the wavelength of the photon is comparable to the atom.

The large atom (144^3 torus, 11.9 million components, m = 0.625, $Z\alpha = 0.3$, Bohr radius 6.0 steps). **All five predictions passed.**

quantity	$2p_{1/2} \to 1s$	$2p_{3/2} \to 1s$
frequency against $(3/8)\, m^*(Z\alpha)^2$	+1.04%	+1.66%
golden-rule rate against the dipole formula	+0.83%	−3.68%
rate against the hydrogenic formula $(2/3)^8\, \alpha\, (Z\alpha)^4\, m^*$	+0.99%	−1.67%
identity between the length and velocity forms	0.999936	0.999933

How to read it. The frequencies of the 2p → 1s jumps are right within 1-2%. The emission rates are within 1.7% of the hydrogenic formula.

- The spectrum: 1s 0.5996603, 2s 0.618515, $2p_{1/2}$ 0.6186051, $2p_{3/2}$ 0.6187224.
- $m^* = m/(1 + rm)$ is the kinetic mass of the Wilson electron.
- The comparison is with the nonrelativistic hydrogenic formula at the same $Z\alpha$ and m^*, not with physical

hydrogen.

10. The moving atomic clock (P23-P25c)

What is measured.

- The clock is the energy difference, at fixed total momentum P, between the centroid of the eight n = 2 states (2s and 2p) and the 1s doublet. At rest the 2p-1s difference is also reported.
- $R = \omega(P)/\omega(0)$: the frequency in motion divided by the one at rest.
- The velocity in the tables is *nominal*: $V = P/\sqrt{((m + M)^2 + P^2)}$, with the free masses.
- For a relativistic composite object $\omega(P) = \omega(0)/\gamma$, that is, $R - 1 = -V^2/2$. It is Einstein's slowdown.
- The frequency of an emitted photon, with the recoil and the direction, is not computed.

The model.

- Electron and nucleus. The nucleus is a scalar particle with the lattice dispersion $\sqrt{(M^2 + 4\Sigma \sin^2(q/2))}$.
- They are reduced to the relative coordinate, at fixed P: m = 0.625, M = 2.2222, Zα = 0.375, 144^3 grid of relative momenta.
- P is not a multiple of 2π/144, so the torus carries *twisted* boundary conditions. It is equivalent to the infinite-volume problem sampled on that grid.
- The interactions: instantaneous Coulomb, and a Breit operator added at first order. That operator is the regularized continuum form: $V_B = (Z\alpha/2)\ \Sigma_{ab}\ \alpha_a\ T_{ab}\ v_{N,b}$ (symmetrized), with $T_{ab} = \delta_{ab}/r_s + r_a\ r_b/r_s^3$, $r_s = \sqrt{(r^2 + 0.25)}$, the Dirac matrix α of the electron and the velocity of the nucleus v_N from its dispersion.
- The magnetic exchange was not derived from the Wilson current on the lattice. For the Wilson electron the free velocity is $\alpha_a \cos k_a + \beta r \sin k_a$, not α.

What is expected in the continuum.

- One looks at the dominant Coulomb binding, at first order in the relativistic corrections and at second order in the velocity of the centre of mass, averaged over the multiplet.
- The kinetic terms give $R - 1 = +(5/6)V^2$.
- The Darwin-Breit term gives $-(4/3)V^2$.
- The total is $-V^2/2$. It is the Breit term that makes the binding energy contribute to inertia (Krajcik and Foldy 1974).
- The value is not exact at Zα = 0.375, with a regularized potential and a finite grid.

The ratio R_B is formed from energies corrected at first order. If the ratio itself is expanded strictly to first order in the Breit shift, a second value is obtained. This too is reported.

test	dispersions	$(R - 1)/V^2$, Coulomb only	Breit contribution	$(R - 1)/V^2$, with Breit	the same, strictly first order
P23-P24	Wilson r = 0.2	+1.125 (V 0.15), +1.164 (V 0.30)	−1.116, −1.079	+0.010, +0.085	−0.037, +0.040
P25b	Wilson r = 0.2 / 0.1 / 0.05	+1.134 / +1.188 / +1.215	−1.131 / −1.161 / −1.176	+0.003 / +0.027 / +0.039	−0.042 / −0.023 / −0.014
P25c	continuum, on the lattice momenta	+0.778	−1.215	−0.438	−0.488
continuum, dominant order	—	+0.833	−1.333	−0.500	−0.500

How to read it. The number to look at is $(R - 1)/V^2$ with Breit. Einstein wants −0.500. On the Wilson lattice it comes out close to zero (from −0.042 to +0.085). With the dispersions of the continuum (P25c) it comes out as −0.438, or −0.488 at strict first order.

At rest the atom is right: frequency 0.984 and binding 0.990 of the values with the reduced mass (P23).

In motion, on the Wilson lattice, it is not.

- The Breit term has nearly the expected size, but the kinetic part is too large.
- The coefficient stays close to zero, and its sign depends on the convention: far from Einstein's −0.5.
- **A registered prediction failed (P25b).** It was predicted that reducing the Wilson term would restore the slowdown. Instead the coefficient does not move towards −0.5 as r decreases.
- This does not show that the Wilson term plays no role.

What P25c shows.

- On the lattice the nucleus has M = 2.2: its *Compton wavelength* is shorter than one step. To move the atom at V = 0.15 a total momentum of 0.43 per step is needed.
- There the lattice dispersions differ from the relativistic ones by terms of the same order as the effect sought.
- So the free dispersions of the continuum are used on the same momenta (electron $\alpha \cdot k + \beta m$, nucleus $\sqrt{(M^2 + q^2)}$), with the same Coulomb, the same Breit operator and the same no-pair method. The coefficient becomes −0.438, or −0.488 strictly at first order, against −0.500.
- This change also makes the kinematics of the electron consistent with the vertex α used in Breit.
- **P25c therefore supports this reading:** the coarse kinematics of the lattice contributes substantially to the missing slowdown.
- **It is not a direct test of the continuum limit of the checkerboard with interaction.** For that one would also need: the current and the magnetic interaction derived on the lattice; control of the no-pair approximation, of the finite-volume and cutoff effects, and of the perturbative errors; masses and momenta of the electron and the nucleus all small in lattice units. This convergence has not been demonstrated here.
- The difference that remains (12% in the unexpanded ratio) is of the order of the other lattice effects at this size. It is an order-of-magnitude statement, not an error budget.

Two technical points.

- **P25 is not concluded.** In P25 (Wilson r = 0.1 and 0.05) and in one run of P23 (lighter nucleus), the free band of the states with a negative-energy electron and a fast nucleus, $-E_e(k) + \varepsilon_N(P - k) - M$, reaches 0.67 for r = 0.1 and 0.97 for r = 0.05. It is above the 1s (0.59). For r = 0.2 and M = 2.2222 its maximum is 0.07. There the program that searches for the eigenvalues on $(H - \sigma)^2$ did not converge: its residual, of the squared operator, stayed near 10^{-5}, with six approximate energies within 2×10^{-4}.
- **The no-pair projection.** P25b and P25c use the no-pair projection with the free projectors $\Lambda_+(k) = (1 + h(k)/E(k))/2$ (Sucher 1980), with the negative-energy subspace shifted above the spectrum. It is a change of the model, not only of the computation method: it removes the Coulomb couplings with the negative-energy states. At r = 0.2, where both methods work, it agrees with P23 within 0.65% in frequency and within 0.01 in $(R - 1)/V^2$. It is a numerical check, not an identity. The residuals and the negative-energy part ($\leq 2.4 \times 10^{-14}$) were checked on the eigenvectors before saving them in single precision for the Breit step.

11. The registered tests

How they are counted. One counts the clauses listed separately in the file of the registered tests. P1, P4, P5 and P11c are each divided into two clauses; two arithmetic items are not counted. A prediction passed only in a check made afterwards is counted as failed. A prediction registered for two runs, of which one did not converge, is counted as incomplete.

tests	subject	passed in full / failed / incomplete
P1-P7	the clock	10 / 3 / 0
P8-P9	the checkerboard in 1D and 3D	10 / 0 / 0
P10-P11	mass, relations among particles	6 / 3 / 0
P12-P15	the hand on the boundary	8 / 4 / 0
P16-P19	the field of MM 2.27	8 / 5 / 0
P20	quantized field and matter, 1+1	2 / 1 / 0

P21-P22	the atom	5 / 4 / 0
P23-P25c	the moving atomic clock	10 / 5 / 6
all		**59 / 25 / 6**

- The six incomplete ones are the four predictions of P25 and P23a-b. P23a-b passed for the heavier nucleus and were not concluded for the lighter one.
- Counting instead the items as they were originally labelled, without dividing them, one gets 56 / 24 / 6.
- These totals describe a set of checks of different kinds. They are not a statistical measure of the model.

The failed predictions, and how to read them.

- **P4 (and its hand estimate) and P5 at v = 0.8.** Thresholds too tight, and an estimate that ignored that the longest chain picks the best path. P5 had about 780 copies; the comparison at equal proper time passed. The overall criterion for P1-P5 was therefore not met (section 3).
- **P10b at θ = 0.01:** derivative step too large. **P10d:** the excess of the inertial mass over the rest mass in 3D is about $4\theta^2/3$, not below θ^2.
- **P11c:** the prediction was 8 Weyl nodes, all with chirality +1 (net +4 at ω = 0). There are instead 16, with net chirality 0. The prediction used the wrong rotation condition.
- **P12a-c:** the detector failed because the grid contained k = 0, where the two boundary modes are degenerate (they have the same energy). All three passed only in a recheck with a shifted grid. **P13a:** the decay is not exactly exponential (R^2 0.996 against 0.999).
- **P16c:** the velocity in an electric field oscillated at switch-on (Zitterbewegung), and this was read as a delay. **P17b-c:** tails of the packets, and a torus too small. **P18c:** the projector V V† used was wrong with degenerate eigenvectors. The corrected projector keeps the neutral packets at 12 within 1.5×10^{-5} at t = 20 and 0.019 at t = 30, still above the registered 10^{-6}. **P19b:** the deviation from the black body changes sign later than predicted.
- **P20c:** the Schwinger mass is not reached at N ≤ 20.
- **P21a-d:** the small atom.
- **P24b, P24d, P25b-b, P25b-c, P25b-d:** no slowdown with the lattice dispersions.

The addition of version 2.40.2 (section 3.1) has its own registered tests: 20 passed, 4 failed, 4 not evaluable or incomplete. The totals above do not change.

12. What is not claimed

- **That some structure of this deposit is new.** Time dilation from random events is causal-set theory. The checkerboard and its 3D form are known Dirac quantum walks. The boundary fermions, the lattice field, the Schwinger model and the atom on a lattice are textbook lattice physics. What is proposed is a sequence motivated by the premises of section 2, with the other assumptions declared there.
- **That the value of any mass is derived.** θ is free; only relations among masses follow.
- **That the checkerboard reproduces the moving atomic clock.** At the spacing computed here it does not. P25c, with the dispersions of the continuum, supports the role of the kinematics of the lattice, but it is not a test of the continuum limit with interaction of the checkerboard (section 10).
- **That the Breit operator used is derived from the field on the lattice.** It is the continuum form, regularized.
- **That quantum electrodynamics is solved.** In 3D the photons enter through emission rates at the lowest order and a Breit shift at first order. In 1+1 finite chains are diagonalized exactly, and the continuum limit is not reached (P20c).
- **That the zero-point energy of the field accounts for dark energy.** With a Planck step it is about 10^{123} times too large.
- **That the lattice step is derived.** It is a scale of the model; the Planck value is an assumption.
- **A curved spacetime.** Here everything is in flat spacetime.
- **That the number of dimensions of space is derived.** Sections 3-10 take three directions of space as given. With the pairwise rule of section 3.1, n counted sides give n − 1 directions of space: three directions correspond to four sides. The number of sides is an input, chosen to match the observed 3+1 spacetime. The framework admits three dimensions, it does not derive them (added in 2.40.2).

13. Relation to prior work

element of this deposit	prior work	what is different here

time dilation from random events (section 3)	Brightwell and Gregory (1991); Bombelli, Henson and Sorkin (2009); Ulam's problem (Hammersley 1972; Vershik and Kerov 1977; Logan and Shepp 1977), with the finite-size correction of Baik, Deift and Johansson (1999); Bondi's k-calculus (1964)	premise A motivates the randomness (independence and uniformity are assumed), and B and C the product rule
proper time as the geometric mean of two counts (section 3)	Knuth and Bahreyni (2014): an observer is a chain of an ordered set of events; an interval is quantified by its projections Δp, Δq on two chains, and the consistent scalar is $\sqrt{(\Delta p\,\Delta q)}$, with $\Delta p\,\Delta q = ((\Delta p + \Delta q)/2)^2 - ((\Delta p - \Delta q)/2)^2$; the Minkowski metric and the Lorentz transformations follow (added in 2.40.2)	the pair is a pair of coprime light sides (premise B) and the copies are random (premise A); the identity between geometric mean and arithmetic mean is the same
more than two sides (section 3.1)	real null frames, used by Zeeman (1964) in the proof that causality implies the Lorentz group; symmetric ones in four dimensions: Derrick (1981), Coll and Morales (1991); in emission (GPS) coordinates it is the coordinate covectors that are null: Rovelli (2002), Coll and Pozo (2006). Knuth (2016, abstract of a talk; full text not consulted) reports that with coordinated chains a consistent quantification exists only for two chains (1+1) or four (3+1)	the sides are independent random counts, and the pairwise rule is tested against the product rule on growing structures; with this rule three sides also give a 2+1 structure, so the setting differs from Knuth's, whose choice of four chains is not addressed here
the checkerboard, 1D (section 4)	Feynman and Hibbs (1965); Jacobson and Schulman (1984); Gaveau, Jacobson, Kac and Schulman (1984); Kac (1974); Bisio, D'Ariano and Tosini (2015)	the turns are changes between the two light sides of the clock of section 3
the 3D Dirac quantum walk, spin ½ (section 4)	Jacobson (1984); Bialynicki-Birula (1994); Meyer (1996); Arrighi, Nesme and Forets (2014); D'Ariano and Perinotti (2014)	a palindromic split-step construction in that family; D'Ariano and Perinotti derive the walk from principles of information processing, here it continues the counting clock
doubling, net chirality zero (section 5)	Nielsen and Ninomiya (1981); Wilson (1977)	a discrete-time automaton, where the hypotheses of the theorem are not those of a static Hamiltonian
boundary fermions and their mass (section 6)	Jackiw and Rebbi (1976), for the localization; Kaplan (1992); Shamir (1993), for the chiral lattice construction	—
gauge field and Maxwell on the lattice (section 7)	Peierls (1933); Yee (1966); Wilson (1974); the equations of MM 2.27	the field is the one of MM 2.27, coupled to the counting clock
Casimir, black body (section 7)	Casimir (1948); Planck (1900)	—
Schwinger model (section 8)	Schwinger (1962); Kogut and Susskind (1975); Banks, Kogut and Susskind (1976); Coleman (1976)	—
spontaneous emission (section 9)	the golden rule (textbook form in Fermi 1950); Bethe and Salpeter (1957)	—
moving bound states, Breit (section 10)	Breit (1929); Darwin (1920); Krajcik and Foldy (1974); no-pair projection: Brown and Ravenhall (1951), Sucher (1980); SLAC derivative: Drell, Weinstein and Yankielowicz (1976)	the diagnosis on the lattice: the slowdown is missing at $M \cdot a \approx 2$ and is nearly present with the dispersions of the continuum

How to read the table. The first column is the piece of the work. The second says who had already done it. The third says what changes here.

A limit of the literature check. The check was made by the author's AI assistant on the works listed here, not on the whole literature. The bibliographic data were not all checked against the primary sources.

How the work was done

The registrations. Every test had its predictions written, with a SHA-256 fingerprint, before the calculation, in the author's working files. The fingerprints do not have a public time stamp. The original registrations, in Italian, are in `documenti/registrations_original/` with the registered fingerprints. `documenti/REGISTERED_TESTS.md` gives abridged English translations and every outcome. The addition of 2.40.2 has its tests in `aggiunta_2.40.2/REGISTERED_TESTS_2402.md`.

The files of the versions.

- Programs, outputs and registrations of the 2.40.2 addition are in `aggiunta_2.40.2/` (ADDITION.md).
- Details, programs and registration of the 2.40.1 correction are in `correzione_2.40.1/CORRECTION.md`. The control with $\theta_0 = 0.1$ of section 6 is also there.

Who did what.

- The author is not a professional physicist. The framework and its premises are his.
- The calculations were done with an AI assistant (Claude). The checks made afterwards are marked as such.
- A second AI assistant (Codex) reviewed the text and the code of the programs of the moving atom. It made independent analytical checks: the continuum coefficients, the negative-energy band, the no-pair operator, the count of the predictions. It did not run the programs or consult external sources. Its corrections are applied.
- For version 2.40.2 the same reviewer (Codex, model gpt-6-luna) read the addition in two rounds, without running the programs. Its corrections are applied.
- No peer review by people is claimed.

The check before deposit. Before deposit the author's assistant re-ran from scratch every program that fits on a laptop, and compared the outputs with the reference files. For the addition of 2.40.2 the check re-ran the algebra, the clock ratios, the containment diagnostic and the 3+1 control. It re-analysed the saved outputs of the rank structures and re-ran one structure and one run of the moving clocks (`aggiunta_2.40.2/risultati/COLLAUDO.txt`). The full set of those runs takes about 70 minutes on two processes and was not repeated.

How to redo the calculations (Python 3 with numpy and scipy).

```
python codice/run_all.py        # every program that runs on a laptop (8 GB), about one hour;
                                # outputs in uscite/, compared with risultati/
python codice/tabelle_aws.py    # recomputes the tables of sections 9-10 from the reference outputs
in risultati/aws/
```

- `run_all.py` copies the programs into `uscite/` and runs them there, because some rechecks read the main program by its file name.
- The running times printed by the programs are masked in the comparison. The JSON files are compared by content, with numbers equal within 10^{-9} relative: the programs that search for the eigenvalues differ in the last digits from one run to another.
- `tabelle_aws.py` rebuilds the tables. It does not re-check every registered condition: for P22 it prints one state per multiplet.

The atoms of sections 9-10 need about 40 GB of memory per run. They were computed on AWS (r7i.4xlarge, 16 vCPU, 128 GB; 25-160 minutes per run). Their programs are in `codice/`, the outputs in `risultati/aws/`. The commands are:

```
python atomo_grande.py 144 0.625 0.2 0.3 <folder>                          # P22
python atomo_nucleo.py 144 0.625 0.2 0.375 2.2222 <V> <folder>             # P23 (V = 0, 0.15, 0.30)
python breit.py <folder> 144 0.625 0.2 0.375 2.2222 <V>                    # P24
python atomo_nucleo_np.py 144 0.625 <r> 0.375 2.2222 <V> <folder>          # P25b (r = 0.2, 0.1,
0.05)
python atomo_nucleo_slac.py 144 0.625 0 0.375 2.2222 <V> <folder>          # P25c (r must be 0)
python breit_slac.py <folder> 144 0.625 0 0.375 2.2222 <V>                 # P25c
```

- The environment variable MM_WORKERS sets the number of FFT threads.
- In `atomo_nucleo_slac.py` the Wilson argument must be 0. A non-zero value would change the reduced mass printed in the diagnostics, without adding a Wilson term.
- The outputs of the tests made before registration (torus 32^3-64^3) are in `risultati/prove/`.
- The addition of 2.40.2 has its programs in `aggiunta_2.40.2/codice/` (sympy is also needed). The commands, about 80 minutes in all on two processes, are in `aggiunta_2.40.2/ADDITION.md`, section 7. `aggiunta_2.40.2/codice/collaudo_2402.py` repeats the check made before deposit.
- The SHA-256 fingerprints are in `IMPRONTE.sha256`. The code comments and the printed labels are in Italian.

The Italian summary. The archived text ends with a *Sintesi in italiano*. All its contents are already in the sections of this version.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a complete Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is

authoritative. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **Boundary.** A surface where the turn changes sign.
- **Checkerboard.** The sum over histories with an imaginary weight for each turn; in 3D, a Dirac quantum walk.
- **Chirality, hand.** Whether a particle is *right-handed* or *left-handed*. A state with a single hand is a Weyl fermion.
- **Copy.** An event of the count (premise A).
- **Dirac quantum walk.** A way of making a particle move on a lattice, in steps, which in the limit of small steps follows the Dirac equation.
- **Dispersion.** The relation between frequency ω and wave number k of a particle or of a wave.
- **Lattice.** A grid of points; here space and time made of steps.
- **Light cone.** The region of spacetime that light leaving an event can reach.
- **Light sides a, b.** The two sides of the light diamond between the start and the end of a clock.
- **Longest chain.** The longest row of events between two given events, each inside the light cone of the previous one.
- **Lorentz factor, γ.** $1/\sqrt{(1 - v^2)}$.
- **No-pair.** The projection on the positive-energy states of the free electron.
- **Pairwise rule.** $Q = \Sigma_{i<j}\, x_i x_j$: each pair of sides is multiplied and the pairs are added; the clock reads $\sqrt{Q}$.
- **Proper time.** The time shown by a clock that travels with an object.
- **R.** $\omega(P)/\omega(0)$: the difference between the n = 2 centroid and the 1s of the moving atom, divided by its value at rest.
- **Sides x_1, …, x_n.** Independent counts of a clock; for n = 2 they are the light sides a, b.
- **Spin.** A quantum property of particles that resembles a small rotation on itself, but is not a little ball that spins.
- **Time dilation.** The slowing of a moving clock: the factor $\sqrt{(1 - v^2)}$.
- **Turn.** A change of light direction along a history; its weight carries the mass.
- **V.** The nominal velocity $P/\sqrt{((m + M)^2 + P^2)}$.
- **Wilson term r.** A turn that depends on the neighbours; it makes the extra species of the lattice heavy.

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